Machine Learning 1 - Cheat Sheet

Multivariate Calculus

Index notation

- $[\mathbf{A}\mathbf{v}]_i = \sum_p \mathbf{A}_{ip}\mathbf{v}_p$
- $\mathbf{v}^{\mathrm{T}}\mathbf{A}\mathbf{x} = \sum_p \sum_q \mathbf{v}_p \mathbf{A}_{pq}\mathbf{x}_q$
- $\mathbf{v}^{\mathrm{T}}\mathbf{x} = \sum_p \mathbf{v}_p \mathbf{x}_p$

Multivariate derivatives

- $\frac{\partial}{\partial \mathbf{w}}\mathbf{w}^{\mathrm{T}}\mathbf{v} = \mathbf{v}^{\mathrm{T}}$
- $\frac{\partial}{\partial \mathbf{w}}\mathbf{w}^{\mathrm{T}}\mathbf{A}\mathbf{w} = \mathbf{w}^{\mathrm{T}}(\mathbf{A} + \mathbf{A}^{\mathrm{T}})$
- $\frac{\partial}{\partial \mathbf{w}}\mathbf{A}\mathbf{w} = \mathbf{A}$

Useful functions

- Kronecker delta: $\delta_{ik} = \begin{cases} 1 & \text{if } i = k \\ 0 & \text{otherwise} \end{cases}$
- Indicator function: $\mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$

Conventions

- Vectors are columns ($\mathbf{x} \in \mathbb{R}^{n \times 1}$)
- If $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$, then $\frac{df}{d\mathbf{x}} \in \mathbb{R}^{m \times n}$

Constrained optimization

Equality constraint

$$\max_{\mathbf{x}} f(\mathbf{x}) \text{ subject to } g(\mathbf{x}) = 0$$

- Lagrangian: $L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda g(\mathbf{x})$.

Inequality constraint

$$\max_{\mathbf{x}} f(\mathbf{x}) \text{ subject to } g(\mathbf{x}) \geq 0$$

- Lagrangian: $L(\mathbf{x}, \mu) = f(\mathbf{x}) + \mu g(\mathbf{x})$.
- Solve $\max_{\mathbf{x}} \min_{\mu} L(\mathbf{x}, \mu)$ subject to KKT cond.:

$$g(\mathbf{x}) \geq 0, \quad \mu \geq 0, \quad \mu g(\mathbf{x}) = 0.$$

Probability distributions

Univariate Gaussian

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Multivariate Gaussian

$$p(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{k/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

Laplace

$$p(x \mid \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right)$$

Binomial

$$p(x \mid n, \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

*Other distributions will be provided if needed.

Kernel methods

Kernels

- Given valid kernels $k_1(\mathbf{x}, \mathbf{x}')$ and $k_2(\mathbf{x}, \mathbf{x}')$, the following new kernels will also be valid:

$c\,k_1(\mathbf{x}, \mathbf{x}')$	$f(\mathbf{x})\,k_1(\mathbf{x}, \mathbf{x}')\,f(\mathbf{x}')$
$q(k_1(\mathbf{x}, \mathbf{x}'))$	$\exp(k_1(\mathbf{x}, \mathbf{x}'))$
$k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}')$	$k_1(\mathbf{x}, \mathbf{x}')\,k_2(\mathbf{x}, \mathbf{x}')$
$k_3(\boldsymbol{\phi}(\mathbf{x}), \boldsymbol{\phi}(\mathbf{x}'))$	$\mathbf{x}^\top A \mathbf{x}'$
$k_a(\mathbf{x}_a, \mathbf{x}'_a) + k_b(\mathbf{x}_b, \mathbf{x}'_b)$	$k_a(\mathbf{x}_a, \mathbf{x}'_a)\,k_b(\mathbf{x}_b, \mathbf{x}'_b)$

where $c > 0$ is a constant, $f(\cdot)$ is any function, $q(\cdot)$ is a polynomial with nonnegative coefficients, $\boldsymbol{\phi}(\mathbf{x}) : \mathbf{x} \to \mathbb{R}^M$, $k_3(\cdot, \cdot)$ is a valid kernel in $\mathbb{R}^M$, and A is symmetric positive semidefinite. For $\mathbf{x} = (\mathbf{x}_a, \mathbf{x}_b)$, k_a and k_b are valid kernel functions over their respective spaces.